

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 10 ~ BookStream #002.

BookStream | Parr, Pezzulo, Friston ~ Chapter 10 ~ BookStream #002.03 | 26:10

okay this video is going to be an
overview of par pulo and friston's 2022
active inference textbook chapter 10
active inference as a unified theory of
sentient Behavior I'll read the quote
and then Ali please feel free to give
your first overview take on the chapter
in general we are least aware of what
our minds do best Marvin Minsky so what
is chapter 10
about

okay so chapter 10 is by far my uh most
favorite chapter in the whole book but
uh uh at the same time uh it differs
from all the other chapters um in the
sense that it tries to connect connect
active inference to so many other
disciplines and also uh provides uh some
kind of unifying themes in order to tie
many of the looser strands uh of the
theoretical work that's been done so far
at least until um um I mean the books
publication so it's a pretty pretty uh
enticing chapter in terms of um I mean
providing uh many promising uh or
introducing some
um future research um trajectories or at
least promising research uh trajectories
for future so uh for anyone who wants to
pursue active influence research I
believe this particular chapter here can
provide uh I mean quite a bit bit of uh
interesting ideas to
pursue yes this will just be a quick
overview they're going to connect a lot

of dots they abstract away from the specific active inference generative models discussed in previous chapters and focus on integrative aspects of the framework which as Ali mentions is one of the most exciting and motivating and inspiring

Parts we're going to find out by the end of the chapter whether we have what we sought after a unified perspective on problems like perception action selection attention and emotion regulation all these topics which are studied with obviously a diversity of methods but also on the theoretical side essentially in commiserate Frameworks and so active inference here is being positioned as a way to integrate across these different cognitive phenomena and on the more historical or philosophical side how established theories like cybernetics idea motor theory of action reinforcement learning and optimal control can also be understood within the integrative stance that active inference

provides section 10.2 wrapping up is like a speedrun on the book itself they Summarize chapter one active inference as a normative approach a chapter with a lot of similarities to chapter 10 chapter two the low road to active inference how basian statistics chapter three the high road to active inference the free energy principle and the imperative to exist chapter four the formal aspects of active inference with a focus on

variational inference on generative models chapter 5 they developed from the abstract perspective in chapter 4 and went into more detail on the process theory about how the principle may be implemented by the brain and they gave a bunch of examples of systems of interest and related work on the Maman nervous system chapter six the recipe to design active inference model chapter six kicks off the second half of the book and takes a pragmatic turn towards Hands-On application Learners building their own generative models along the way chapters seven and eight are very curtly summarized here this pair of chapters describes the continuous in chapter eight and discreet in chapter seven generative models and chapter nine on empirical modelbased analysis looks at how to connect data to generative models so any comments on that or continue on with section 10.3 connecting the dots I think we can go on to chapter to section 10.3 because uh you provided the basic summary of section 10.2 perfectly thanks so what about 10 three okay so 10.3 um is about going Beyond some uh specific uh aspects of um I mean cognitive facilities of of the agents and try to uh somehow uh integratively model uh the different sub systems of um cognitive facil of the of uh the particular agents um cognition uh so for for instance we can model memory attention perception and uh all all of these uh different aspects of cognition

um in this integrated uh formula
of active inference within this
integrated formalism of active inference
and try to uh somehow see the
relationships between all of these uh
different aspects of cognition and I
believe uh this uh integrative work can
provide um many
illuminating uh insights about how
various aspects of uh given agents
cognition is related to each other which
is I believe a very important open
question in cognitive psychology as
well great and the first paragraph
references
this direction in cognitive science
which is rather than to develop a simple
or Advanced model of one cognitive
phenomena separated out from the others
the suggestion of Daniel dennet in 1978
was to model a whole iguana a complete
cognitive creature perhaps a simple one
but 103 reiterates the way that active
inference can be used to connect dots
and integrate across cognitive phenomena
okay 104 predictive brains predictive
minds and predictive processing what
would you say about
this uh all right so this is also a
recurring question uh among the people
who uh who were introduced to active
inference for the very first time um
namely what is uh the distinction
between active inference and all the
other uh related theoretical Frameworks
such as predictive uh predictive
processing predictive coding and so on
and uh this part this section uh gives

some hint about uh the similarities and also the differences uh between all of these uh different approaches and how they relate to each other whether they're subset um subset of one one of each other or they're distinctly different uh and fundamentally different from each other so um this this section here

10.4 um I believe can be helpful to anyone who wants to understand how active inference fits into this um General framework of recent predicted um predictive way of doing uh cognitive psychology

research great and one nice quote here active inference creatures fulfill two imperatives epistemic and pragmatic that's in reference to the expected free

energy all right Section 105

perception okay so if we remember from uh chapter 10 uh we were introduced to active inference first by observing how it can model uh the phenomenon uh perception of of an agent um as opposed to uh modeling its action and then we saw that they're basically uh I mean uh they mirror uh image of each other uh the equations that describe those uh phenomenon uh are basically the same equations but

just flipped for um the situation we want to model and

here uh there's this generalization of how to um how how does it mean to model perception as opposed to uh action and in particular uh what's the basian brain

uh hypothesis um I mean what is uh the particular approach taken by basian brain hypothesis uh for modeling uh perception as opposed to some uh other conventional um approaches um to model the same

phenomena great 105 perception flows into 10 six action control so these two sections

describe the perception and action view of active inference in this action Theory there's subsections on idea motor theory which is a theory with a long history that relates to how cognition influences action cybernetics active inference is closely related to cybernetic

ideas and optimal control theory another framework for thinking about action

selection these sections differentiate some distinctions of active inference and optimal control theory cybernetics and IDE motor theory however largely as their systems of Interest tend to be overlapping there's more in common with these approaches that students of them will find quite familiar in active inference but again the authors take care to distinguish what active inference does differently than these approaches as well anything else to add there um nothing in particular thanks all right 107 utility in decision-making action expresses priorities

Mahatma

Gandhi uh okay so here in this section there's also uh this kind of comparative

analysis between uh active inference

Theory and uh many

various a few U different approaches uh

used in machine learning so for instance

uh we have uh reinforcement learning

uh and um I mean various ways of doing

um modeling the planning and decision

making such as uh the general basian

decision Theory uh which is a super set

of many different approaches used to

model uh the decision making process

using basian inference uh and um also

there's this planning as inference which

is something uh that active inference

emphasizes uh uh as again as opposed to

some other traditional or conventional

uh way of modeling decision making

because

here as it I mean it's not just simple

planning uh as a consequence of uh

inference but uh but I mean planning

itself is uh considered um I mean the

inference uh considered itself as way of

uh constructing the U expected free

energy uh so uh in this way we can

regard planning as inference itself so

um this is one of the Novelties of

active

inference uh compared to any other

approaches as also explicitly stated on

page

2011

all right 108 behavior and bounded

rationality okay so uh this section

also uh points to some interesting work

um with regards to how we can model um

different conceptions of rationality and

specifically uh bounded rationality and

um there are some interesting papers uh
which um came out after the publication
of uh this book uh in
which some some researchers uh pursue
this line of research um in order to U
model uh even the cons the way people
believe conspiracy theories or the way
people

uh do or make uh apparently irrational
decisions and uh they manifest some
irrational behavior and so on uh so uh
this

section points out uh to some tentative
ways that active inference can be
employed to model uh rationality as well
so it's not just about uh modeling
perception and action which is a
wellestablished way of uh doing AC
inference using active inference but
from this section onwards uh we uh
gradually move towards some Uncharted
territories uh which uh I mean
admittedly some of them are still uh at
their very nent stages

so

awesome section 109 veilance emotion and
motivation

okay again uh as I already mentioned uh
this is also one of the recent
developments of active inference how to
model uh emotion with act within active
inference framework so I believe uh one
of the uh earlier one of the earliest
papers that um did this kind of modeling
was uh

I forgot the title

but it's a way of um um of modeling uh
an a an agent's behavior in terms of uh

the positive or negative valence uh and
uh which directly reflects the emotional
state of that agent so by agent uh it
doesn't necessarily mean a human agent
but we can see that this kind of balance
in terms of positive and negative
balance and how it can uh motivate the
behavior of the agent uh can be captured
with uh within active inference
framework
using um using the same techniques we've
seen uh for epistemic foraging and so on
so uh it's an interesting way of
connecting
those two
subdisciplines yeah I think it's deeply
felt affect there's other related words
yes that was
it again another sort of complex or
composite life phenomena moving from
emotion onto homeostasis allostasis
interceptive processing this section
conveys work from variety of authors
that's been continued on about
homeostasis and allostasis which is the
anticipatory steering towards preferred
States and section 1011 attention
salience and epistemic Dynamics again
approaching a sort of cognitive
phenomena
Nexus and differentiating similar with
rule learning causal inference and fast
generalization here moving slightly more
towards the computational cognitive
side anything to say on those sections
1012 1011 or
1010 uh I just wanted to point to uh the
way that uh some so up to this point

we've seen uh mostly uh the application oriented research of active inference uh in order to deal with u various sorts of scenarios uh I mean from modeling uh the very basic uh cognition of um sentient agents to uh modeling some even social and cultural phenomena uh but I just wanted to point to some uh interesting theoretical work uh that's been going on in parallel with these uh application oriented Works uh which I believe can open up even uh more uh opportunities and uh probably even

uh I mean

drastically change the view of active inference um in terms of how to how to approach uh different scenarios and different uh phenomena um so uh yeah this kind of philosophical and theoretical uh aspect active inference uh

although it's still in its early stages um is beginning to uh lead to some very interesting results uh in terms of applicability of active

inference awesome and then section 1013 active inference and other fields open directions as the title would seem to set off this section

goes into some of the extensions and elaborations that are quite ongoing in active inference especially related to Social and cultural Dynamics and machine learning and Robotics so subsection one social cultural Dynamics subsection two machine learning and Robotics anything to say on that section uh nothing other than uh there

are lots of uh interesting uh um
interesting development in this in these
uh particular domains uh especially
after uh I mean the explosion of llms
and generative AI uh we can see the
application of uh active inference even
to uh these Cutting Edge u u
technological developments as
well oh and then section 1014 summary
over read the Lord of the Rings quote
and then look forward to your take on
this last section home is behind the
world ahead and there are many paths to
tread through Shadows to the edge of
night until the stars are all a
light great so uh obviously this is the
last chapter of the whole book and uh
personally for me uh reading through
this book uh I mean reading reading it
multiple times uh have always been uh I
mean an increasingly enjoyable Journey
for me so each time uh I look at I mean
different sections and different uh
chapters of the book uh I I feel that
there are lots of um interesting
potentialities uh that can be uh pursued
further to uh and probably perhaps even
lead to some uh I don't know some
spin-off
theories and um some uh Branch to some
interesting new Fields uh that is
related to active inference but uh they
somehow begin to follow the life of
their own so uh the uh the main uh
strength of this book in my opinion is
exactly in providing uh these uh kind of
Unsettled ideas to develop further uh and
uh this is in my opinion uh one of the

main indicators of a great book uh which uh I mean much more than trying to answer the questions it provides some fertile questions uh in the mind of the reader great points totally concur time and time again we've delighted in this work it's a two paragraph section the first paragraph is worth reading from we started this book by asking whether it is possible to understand brain and behavior from first principles we then introduced active inference as a candidate Theory to meet this challenge we hope that the reader has been convinced that the answer to our original question whether it's possible to understand brain and behavior from first principles is yes we know it side there

on how do you feel and then the second paragraph is a quite direct perforation of the fourth

wall and they make a really pragmatic day-to-day point about our lives as people learning and applying active inference which is it is important to emphasize that active inference is not something that can be learned purely in theory we encourage anyone who has enjoyed this book to think about pursuing it in

practice important writes a passage in theoretical neurobiology and eventually so many other fields too are trying to write down a generative model experiencing the frustration when simulations misbehave and learning from violations of your prior belief when

something unexpected happens and this is something that comes up all the time there's so many incredible highflying philosophical questions and in the abstract quite philosophical they are however once there's a generative model on the table once chapter six has been engaged with a lot of times the abstractions and generalizations are left out and you can actually say well how is attention being dealt with in this generative model or what other way could it be done and so making the generative model is a huge moment and they emphasize that that's how you can learn by doing but it's not the only way that you can pursue and engage active inference day to day it could be in the time interval that's subliminal and between iades it could be having what you expect and prefer for dinner it could be allostasis embedded extended enacted and cultured allostasis and the authors are confident which is to say precise in their beliefs that we will continue to pursue active inference in some form any closing thoughts on 10

ali uh I just wanted to attest um to the last sentence of the book uh because uh definitely after U getting uh familiar uh more and more with different aspects of active inference I can see it um I mean realized or uh or to uh to say it in a better terms I can see it relevant uh to many many different aspects of life not just uh empirical or scientific research but every aspect of

life awesome all right well thank you
for watching this overview video or
playlist and we hope to see you engaged
in a textbook group thank
you
thanks